



Characteristics and dynamics of *Salmonella* diversity and prevalence of biomarker genes in Port Blair Bays, South Andaman, India

Balakrishnan Meena^{a,1,*}, Lawrance Anburajan^{a,1}, Karuppaiya Selvaganapathi^c, Nambali Valsalan Vinithkumar^a, Gopal Dharani^b

^a Atal Centre for Ocean Science and Technology for Islands, National Institute of Ocean Technology, Ministry of Earth Sciences, Government of India, Port Blair 744103, Andaman and Nicobar Islands, India

^b Marine Biotechnology Division, Ocean Science and Technology for Islands Group, National Institute of Ocean Technology, Ministry of Earth Sciences, Government of India, Chennai 600100, Tamil Nadu, India

^c Department of Marine Biotechnology, Bharathidasan University, Tiruchirappalli 620024, Tamil Nadu, India

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ABSTRACT

Salmonella is a major cause of morbidity and mortality in humans worldwide, and the infection with multidrug-resistant strains can cause severe diseases. Many coastal cities around the world discharge their wastewaters into the marine environment. These wastewaters contain a variety of pathogenic microorganisms that may have a role in the contamination of this ecosystem and have potential risks for public health. Using an environmental approach, the present study investigated the presence of *Salmonella* in sediment and water samples collected from Port Blair Bays. In this environmental approach, the provided information about the diversity of the *Salmonella* serovars, antibiotic resistance and the prevalence of virulence factors in *Salmonella*, especially from the coastal waters of Port Blair Bays. The occurrence of *Salmonellae* was significantly higher in water column samples (2.9%) than in those taken from the marine sediments (0.7%). Of the 133 positive *Salmonella* strains, 22 different serovars were identified. *Salmonella enterica* serovar Senftenberg was the predominant serovar, being represented by 54 isolates (42.5%), followed by serovar Typhimurium (19 isolates [15%]) and serovar Agona (12 isolates [9.4%]). The presence of virulence genes (*filC*, *sitC*, *hilA*, *invA*, *sipC*, *hliD*, *hliC*, *invF*, *invE*, *invH*, *sipF*, *aadA*, *pare*, *gyrA*, *spaP* and *parC*) and susceptibility studies with 10 selected antibiotics were also performed. The results of this study revealed that all *Salmonella* isolates were positive for targeted virulence genes and were resistant to at least one antibiotic. Antibiotic susceptibility studies revealed the presence of multidrug resistant *Salmonella* strains in coastal water, which usually from land base sources end up in the marine environment and may pose a significant risk on public health.

1. Introduction

Salmonella are facultative anaerobes from the *Enterobacteriaceae* family which do not develop spores and are Gram-negative rod-shaped gamma-proteobacteria as well as parasites commonly occupying the intestinal zones of humans, cattle, avian, insects and reptiles (Andino and Hanning, 2015). These Gram-negative rods are the etiologic agents of food-borne salmonellosis and also the agents that cause typhoid and paratyphoid fevers. Although food products, including shellfish, are the most common sources of salmonellosis, *Salmonella* is a prime example of a water- and shellfish-transmitted pathogen. *Salmonella* is a large genus of bacteria including more than 2500 serotypes, and diagnosis in the majority of laboratories rely on costly, laborious culture screening

with both nonselective and selective media (Kilger and Grimont, 1993). Both human and animal excreta are sources of *Salmonella*, and many potential routes exist for the transmission of these excreted enteric pathogens. *Salmonella* bacteria display a high degree of resistance to a large variety of stress factors, which provides them with an enhanced capacity to persist in changing environments (Winfield and Groisman, 2003). However, the persistence of organism outside of the host is not uniform among different serogroups (Bradd et al., 2009). The reported different serotypes of *Salmonella* included in the current classification scheme (Popoff et al., 2004) are dominantly identified from human and animal sources (Uzzau et al., 2000).

Information about the groups that predominate in a given environment and their relationship to potential human or animal origins

* Corresponding author at: Atal Centre for Ocean Science and Technology for Islands, National Institute of Ocean Technology, Ministry of Earth Sciences, Government of India, Dollygunj P.O., Port Blair 744 103, Andaman and Nicobar Islands, India.

E-mail address: meenasteffy@gmail.com (B. Meena).

¹ These authors contributed equally to this work.

remains scarce. Nevertheless, the survival capacity may depend on species and pollution sources. Although most studies have focused on the determination of *Salmonella* strain concentrations in some polluted areas, it was recently shown that the annual bacterial loads of this pathogen in rivers and coastal areas can be very important (Baudart et al., 2000a). However, information on the diversity and occurrence of *Salmonella* strains is very scarce (Baudart et al., 2000b), and as a consequence, the ecology of these species remains unknown. This is partly due to the laborious methods required for the detection, isolation, and identification of *Salmonella* strains (Venkateswaran et al., 1989).

The emergence of antimicrobial resistance in *Salmonella* strains is a serious health problem worldwide (Chiu et al., 2002). In the early 1960s, the first incidence of *Salmonella* resistance to a single antibiotic, namely chloramphenicol, was reported (Montville and Matthews, 2008). Since then, the frequency of isolation of *Salmonella* strains with resistance towards one or more antimicrobial agents has increased in many countries, including the USA, UK and Saudi Arabia (Kqueen et al., 2008). Antimicrobial agents such as ampicillin, chloramphenicol and trimethoprim-sulfamethoxazole are used as the traditional first line treatments for *Salmonella* infections. *Salmonella* spp. resistant towards these agents are referred to as multi-drug resistant (MDR). For many years, the phenotypic trait of MDR was widely distributed among *S. typhi* and at a lower rate, among *S. paratyphi* (Rowe et al., 1997). Africa and Asia are two continents with a high isolation frequency of *S. typhi* displaying MDR phenotype. In a surveillance study conducted in five Asian countries, India, Pakistan and Vietnam had higher rates of MDR isolates of *S. typhi* than Indonesia and China (Ochiai et al., 2008).

Salmonella express a variety of virulence factors which mediate the organism's pathogenicity. These factors may include polymorphic surface carbohydrates, multiple fimbrial adhesins, phase-variable flagella and well-structured mechanisms for invasion and survival in host macrophages and other cells. There are approximately 200 genes including those on the five chromosomal pathogenicity islands (SPI-1 to SPI-5) on *Salmonella* chromosomes that are essential for virulence (Hensel, 2004). In vitro characterization of *hilA*, *hilC*, *hilD* and *invF* suggests that these regulators operate in a hierarchy to modulate invasion gene expression with *hilA* acting as the central regulator in this cascade. The *hilA*, is required for expression of SPI-1 invasion genes (Schechter and Lee, 2000). The *invA* gene codes for proteins in inner membranes of bacteria which are necessary for invasion of epithelial cells (Darwin and Miller, 1999). The *invF* and its putative chaperone protein, *sicA* (along with *hilA*) is responsible for activating and controlling the expression of TTS-associated proteins encoded within and outside of SPI-1 (Darwin and Miller, 2000). The secreted effector *sipA* together with *sopA*, *sopB*, *sopD*, and *sopE* act in concert to induce diarrhoea (Zhang et al., 2003), whereas the *sipB* protein has been identified as the bacterial effector responsible for induction of apoptosis. The *sipC* acts as a translocase, mediating bacterial entry into epithelial cells and is also translocated into the host cytosol (Hayward and Koronakis, 1999). It has been suggested that *sipA*, along with the proteins *sipB* and *sipC* form a translocation complex that delivers effector proteins into the host cells (Zhang et al., 2003).

Wastewater is known to be a common vehicle for the transport and transmission of *Salmonella* serovars and is able to pollute environment and infect humans. Indeed, several outbreaks and contaminations have been related to *Salmonella* in wastewater through irrigation of crops (Melloul and Hassani, 1999; Heaton and Jones, 2008), infiltration and transport in soil to groundwater or to its discharge into the marine environment (Harakeh et al., 2006). Although some pathogens are naturally present in the aquatic environment, *Salmonella* may be introduced through animal or human faecal and sewage pollution (Amagliani et al., 2012). The risk of contamination linked to *Salmonella* is enhanced by its ability to survive longer in the environment. Indeed, contamination of groundwater, environmental water and soil by *Salmonella* is due to its ability to survive very long time in these environments (Winfield and Groisman, 2003). Furthermore, contaminations related to *Salmonella* in

seawater and seafood is increased by the capacity of *Salmonella* to survive in relatively high salt conditions (Johnson et al., 2003). Thus, the persistence of wastewater discharges in coastal waters may increase the load of *Salmonella* in marine environment, and therefore increase the higher incidence of seawater and seafood borne infections. As reported, wastewater is an ideal source of *Salmonella* serotypes, in coastal cities and the marine environment is usually the final end of treated and untreated sewage. As reported in wastewater, data from marine environment also reveal a large diversity of *Salmonella* serotypes.

The *Salmonella* spp. is one of the major pathogenic bacteria that could be conveyed to the environmental waters through sewage pollution. Therefore, determination of the presence of *Salmonella*, its levels and its antibiotic resistance in wastewaters and environmental waters are crucial and are required to assess the related health risks (Salem et al., 2017). Pathogenic bacteria become normally more harmful to environment and humans when they acquire resistance to antibiotics. In Andaman and Nicobar Islands, *Salmonella* in wastewater and its influence in marine environment is rarely studied and its bibliography is very scarce. In this context, we are currently leading the pioneering study about *Salmonella* in the southern region of Andaman Islands. This work is focusing on the identification, serotyping, antibiotic susceptibility and prevalence of virulence factors of *Salmonella* strains isolated from Port Blair Bays of South Andaman.

2. Materials and methods

2.1. Study area

Port Blair Bay, located in South Andaman Island extends as a narrow stretch in northeast to southwest direction and opens to Andaman Sea on the eastern side (Fig. 1). The bay is having patches of mangrove plants. Depth of the bay ranges 3 to 25 m whereas in open sea area the depth is more than 50 m. This Bay area is affected by the major anthropogenic activities of Port Blair city, which congregate major share of the total population of Andaman & Nicobar Islands. The sea level in Port Blair region has risen from the previous level due to the land subsidence during the 26th December 2004 M9 earthquake followed by Tsunami. Large area of coastal land has been inundated during Tsunami and a considerable area of coastal land has been permanently submerged due to land subsidence which changed the previous intertidal sea water level (Cho et al., 2008). Due to coastal land submergence, the major portion of previous inter tidal region of coastal area has been lost and a considerable area of mangrove forest has been degraded (Cho et al., 2008).

The climate of these Islands is typically tropical, with heavy gales, cyclones and hot and humid conditions. The annual temperature ranges from 22.5 °C to 35.5 °C, with slight increase from north Andaman to south Nicobar Islands. The relative humidity of ambient atmosphere reaches about 80%. In the present study, five stations were selected in Port Blair Bay from inner bay to open sea namely Flat Bay (FB), Minnie Bay (MB), Junglighat Bay (JB), Phoenix Bay and Aberdeen Bay (AB) (Fig. 1).

2.2. Sample collection and processing

Water and sediment samples were collected during low and high tide from five selected sampling stations along the Port Blair Bays during January 2019. Samples were collected, from 0.3 m below the surface water by using Niskin Go-Flo water sampler and were transferred in sterile screw-top bottles according to the sampling procedures in the USEPA microbiology methods manual (USEPA, 2003). Water temperature and pH were determined in situ using a calibrated thermometer with ± 0.1 °C accuracy (Brannan, UK) and a pH meter (Thermo Orion 420 A plus, USA), respectively. The samples were transported in an ice box to the laboratory and processed within 4 h of collection for the presence of *Salmonella* sp. Spread and membrane filtration methodology was followed for the isolation and enumeration of *Salmonella* isolates. The samples were mixed thoroughly and undiluted

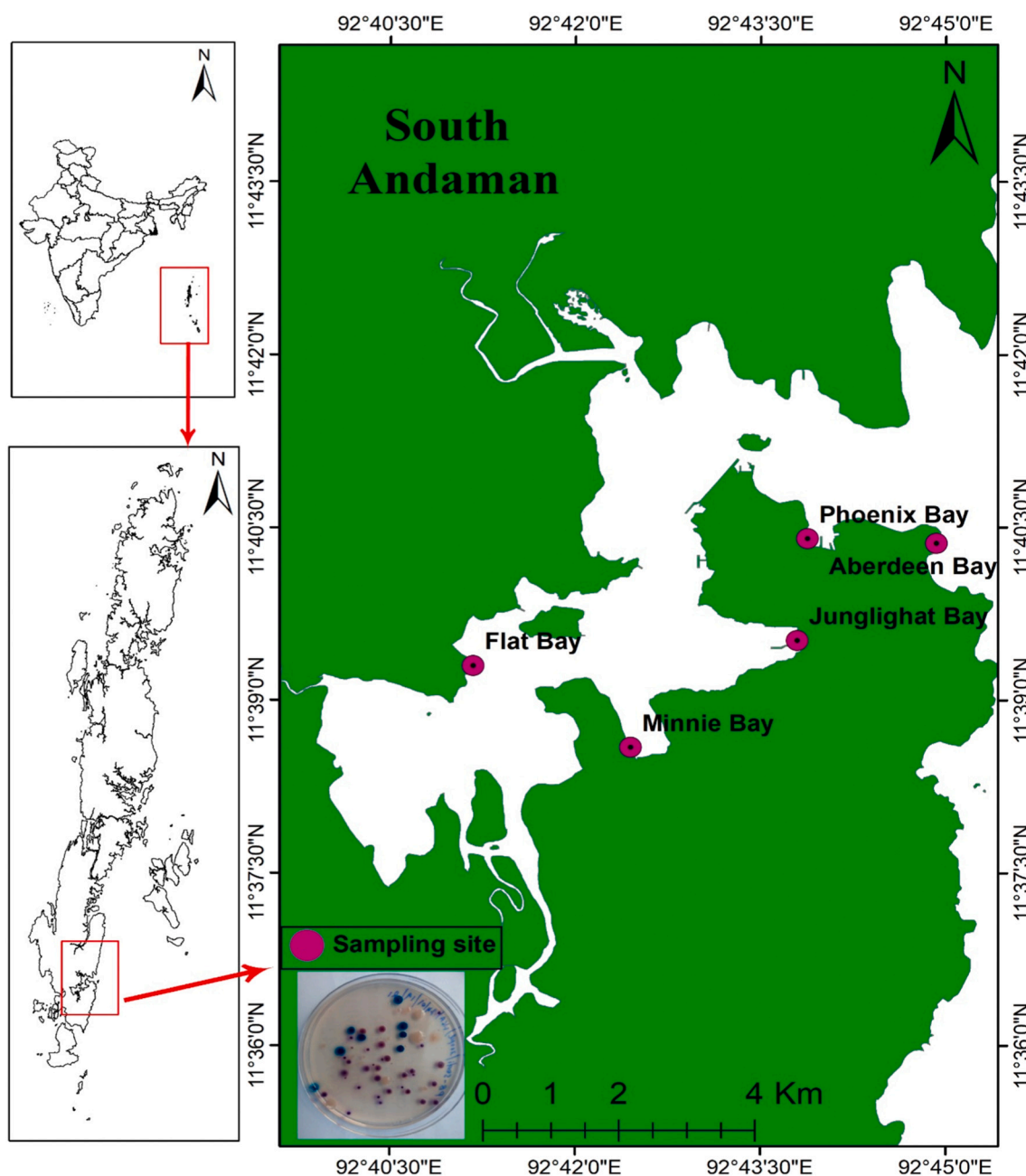


Fig. 1. Sampling stations along the Port Blair bays, Andaman & Nicobar Islands.

samples were filtered through 0.22 μm membrane filters (Millipore Corporation, Bedford, MA, USA) and the membrane filters were placed on pre prepared HiCrome *Salmonella* Agar (HiMedia, India). Triplicate analysis were performed for each samples and *S. enterica* ATCC35664 and *E. coli* ATCC25922 were used as positive and negative controls respectively to validate the HiCrome *Salmonella* agar. Colonies producing dark purple colour were considered presumptive for *Salmonella* species and counted, the result expressed as CFU/ml for water and CFU/g for sediment samples.

2.3. Biochemical characterization and identification of *Salmonella* isolates

Typical dark purple colour presumptive *Salmonella* colonies from the HiCrome *Salmonella* agar plates were selected for further characterization. *Salmonella* isolates were confirmed by selective plating in various selective agar viz., MacConkey agar, Brilliant green agar (BGA), Bismuth sulphite agar (BSA), Hekton's enteric agar (HEA), Xylose lysine decarboxylate Agar

(XLD) and biochemical tests viz., Gram staining, motility, triple sugar iron test, growth on lysine iron agar, indole, MR-VP, urease, citrate utilization, utilization of glucose, lactose, sucrose and dulcitol (Nair et al., 2015). Ability of the *Salmonella* isolates to grow at different temperature at 10 °C, 20 °C, 30 °C, 40 °C, 50 °C, 60 °C, 70 °C and 80 °C were determined (Williams et al., 1988) and the results were documented after 18–24 h of incubation. All the *Salmonella* isolates were subjected for the sodium chloride tolerance test at various concentrations 0.0%, 1.0%, 2.0%, 3.0%, 4.0%, 5.0%, 6.0%, 7.0%, 8.0%, 9.0% and 10.0% (w/v) NaCl. The pH tolerance assay with the different range of pH in the culture medium (4, 5, 6, 7, 8, 9, 10, 11, 12, 13 and 14) and incubated at 37 °C for 24 h. The results were compared to the reference strain of *S. enterica* ATCC35664.

2.4. Screening of protease, lipase and gelatinase activity

Protease activity in *Salmonella* strains was examined by using the single-diffusion method on 1.0% (w/v) nutrient agar medium,

Table 1
Polymerase Chain Reaction assay conditions used in this study.

Biomarker genes	Reaction mixture	PCR conditions	No. of cycles	Reference
<i>filC</i> , <i>invF</i> and <i>invE</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 38 °C, 1 min 72 °C, 30 s	30	This study
<i>sitC</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 41 °C, 1 min 72 °C, 30 s	30	This study
<i>hilA</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 35 °C, 1 min 72 °C, 30 s	30	This study
<i>invA</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 36 °C, 1 min 72 °C, 30 s	30	This study
<i>sipC</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 35 °C, 1 min 72 °C, 30 s	30	This study
<i>hilD</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 35 °C, 1 min 72 °C, 30 s	30	This study
<i>hilC</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 42 °C, 1 min 72 °C, 30 s	30	This study
<i>invH</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 32 °C, 1 min 72 °C, 30 s	30	This study
<i>sipF</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 40 °C, 1 min 72 °C, 30 s	30	This study
<i>aadA</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 45 °C, 1 min 72 °C, 30 s	30	This study
<i>parE</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 42 °C, 1 min 72 °C, 30 s	30	This study
<i>gyrA</i> and <i>spaP</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 46 °C, 1 min 72 °C, 30 s	30	This study
<i>parC</i>	1 μ L dNTP's (2.5 mM), 2 μ L of each primer (20 pmol), 1 U <i>Taq</i> polymerase, 1.5 μ L $MgCl_2$ (2 mM), 1 μ L DNA	94 °C, 1 min 43 °C, 1 min 72 °C, 30 s	30	This study

supplemented with 1.5% of skimmed milk (Iyer et al., 2000). In brief, a straight line at the centre of the nutrient agar plate was inoculated with each test strain. After incubation at 30 °C for 24 h, the test was considered as positive when an opaque halo around the growth was present. Lipase was examined by using the single-diffusion method on nutrient agar plate containing 10 g, NaCl; 10 g, peptone, 0.1 g, anhydrous $CaCl_2$; 20 g, agar; 10 mL, Tween 80 and 1 L distilled H_2O . After incubation, the presence of clear halos around the colonies was considered as positive. Gelatinase was tested using the single-diffusion method on nutrient agar plate containing 3.0%, NaCl and 1.5%, gelatine (Elliot et al., 1995).

2.5. 16S rDNA sequence analysis and molecular identification

Genomic DNA was extracted from *Salmonella* strains and 16S rDNA was amplified using polymerase chain reaction (PCR) method. The universal 16S rDNA eubacterial primers 16S f (5'-ACTCAAGGAATTG ACGG-3') and 16S r (5'-TACGGCTACCTGTGTACGACTT - 3') were used for PCR amplification. PCR amplification was performed with the following conditions: 30 cycles at 95 °C for 1 min, 58 °C for 1 min and 72 °C for 1 min, followed by a final extension step at 72 °C for 5 min. PCR products were analysed in 1.0% agarose gel and the amplicon was cloned in a T/A cloning vector, according to the manufacturer's instructions provided in the InsTAclone PCR Cloning Kit (MBI Fermentas, USA). The acquired 16S rDNA sequences were used in a homology search of the available sequences in GenBank using the BLAST program of NCBI (<http://www.ncbi.nlm.nih.gov>) to determine pairwise identities. Multiple sequence alignments of the sequences were performed

using CLUSTAL-X version 1.81 program, and a phylogenetic tree was constructed with MEGA v5.0 using the neighbor-joining method.

2.6. Antibiotic susceptibility test for *Salmonella* isolates

Susceptibility to several antibiotic agents was determined using a disk-diffusion technique (CLSI, 2009). Bacterial suspensions were spread on Muller-Hinton agar (HiMedia, India) plates, and the antibiotic discs were placed and incubated at 37 °C for 18–24 h. The diameter of the antibiotic inhibition zone was measured in millimetres and recorded according to CLSI M02-A10: susceptible (S), intermediate resistant (IR) or resistant (R). The following antibiotics, with their concentrations given in parentheses, were tested: penicillin-G (P; 10 μ g), kanamycin (K; 30 μ g), ampicillin (AM; 10 μ g), streptomycin (S; 10 μ g), erythromycin (E; 15 μ g), gentamycin (GM; 10 μ g), clindamycin (CD; 2 μ g), chloramphenicol (C; 30 μ g), rifampicin (RA; 5 μ g), oxacillin (OX; 1 μ g). *S. enterica* ATCC35664 and *E. coli* MTCC443 were used as positive and negative control strains, respectively. The selected antimicrobials were chosen based on their common use in treating or preventing *Salmonella* infection in humans. The MAR index was determined by taking the ratio between the number of antibiotics to which the organism was resistant and the total number of antibiotics used (Kaspar et al., 1990).

2.7. PCR amplification of biomarker genes

The biomarker genes; *filC*, *sitC*, *hila*, *invA*, *sipC*, *hilD*, *hilC*, *invF*, *invE*, *invH*, *sipF*, *aadA*, *parE*, *gyrA*, *spaP* and *parC* were PCR amplified using gene-specific primers. PCR was performed in a 25 μ L reaction mixture

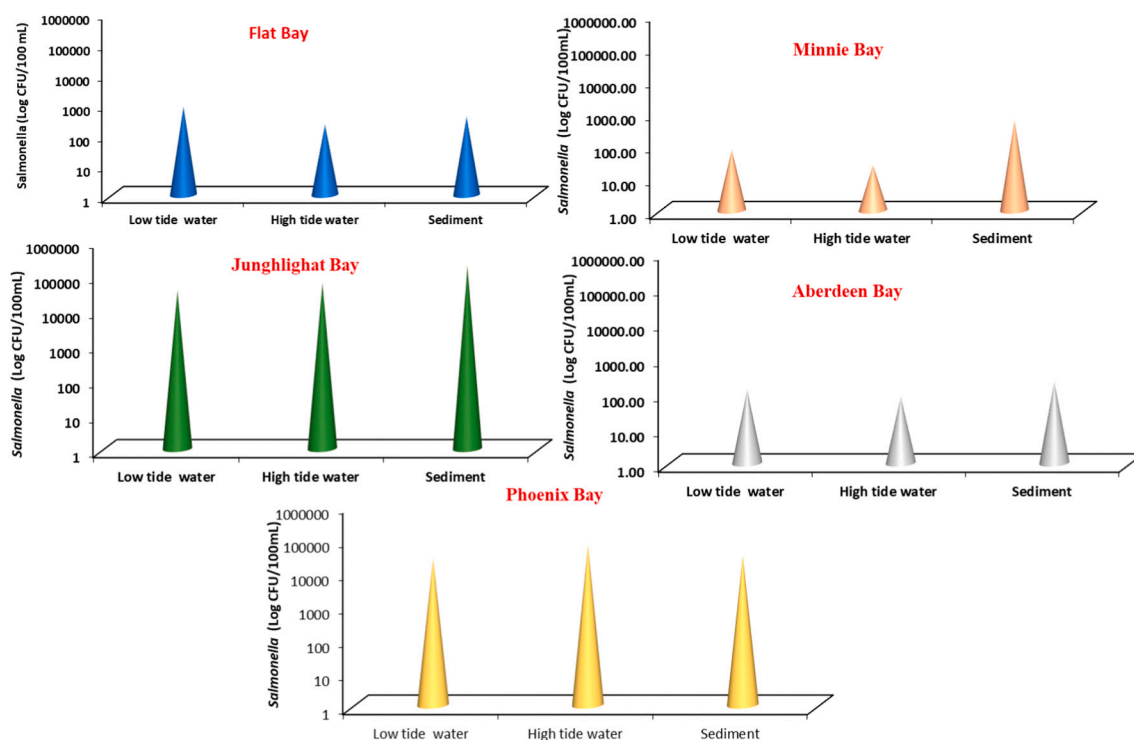


Fig. 2. Population density of *Salmonella* isolates in Port Blair Bays.

that contained 50 ng of genomic DNA, 0.5 μ M of each primer, 200 μ M of each dNTP (MBI Fermentas), 1.25 U of *Taq* DNA polymerase (MBI Fermentas), 1 $\times$ *Taq* buffer, 2.0 mM $MgCl_2$ and autoclaved Millipore water. Amplification was performed in a Mastercycler (Eppendorf) with repeated cycles of 94 $^{\circ}C$ for 30 s, 50 $^{\circ}C$ for 1 min and 72 $^{\circ}C$ for 2 min, and a final extension at 72 $^{\circ}C$ for 10 min. The cycling conditions and reaction mixtures for all biomarker genes are described in Table 1. PCR amplicons were analysed on a 1.5% agarose gel along with a DNA molecular weight marker (MBI Fermentas) and documented with a gel documentation system (UVP BioSpectrum Imaging system, USA). Each PCR assay was performed with *S. enterica* ATCC35664 as a positive control. Positive amplicons as determined by size were purified using a MinElute Gel Purification Kit (Qiagen, Germany).

2.8. Molecular cloning and sequencing of biomarker genes

Gel-eluted PCR amplicons of the biomarker genes were cloned in a T/A cloning vector according to the manufacturer's instructions provided in an InsTAclone PCR Cloning Kit (MBI Fermentas). pTZ57R/T-biomarker gene constructs were transformed into competent *E. coli* JM109 (recA1, endA1, gyrA96, thi-1, hsdR17 (rK-mk+), e14-(mcrA-), supE44, relA1, and Δ (lac-proAB)/F0 (traD36, proAB+, lac Iq, lacZ DM15), plated on Luria-Bertani (LB) agar containing 100 μ g/mL ampicillin, 50 μ M isopropyl-b-D-thiogalactoside (IPTG) and 80 μ g/mL Xgal and incubated overnight at 37 $^{\circ}C$. White colonies were selected for PCR amplification with vector primers M13f-M13r (MBI Fermentas), and clones with the correct insert as determined by size were sequenced with an ABI PRISM 377 genetic analyzer (Applied Biosystems).

2.9. In silico sequence analysis of biomarker genes

The obtained nucleotide sequences were compared with database sequences using the BLAST program of NCBI (<http://www.ncbi.nlm.nih.gov>) and were aligned and clustered using CLUSTAL-X program. The output alignments were imported into GeneDoc program (<http://www.psc.edu/biomed/genedoc/>) and BioEdit version 7.05 (www.mbio.ncsu.edu/BioEdit/) to calculate the percent identities among the nucleotide

and amino acid sequences. Phylogenetic tree was constructed using MEGA version 5.0 by the neighbor-joining method. Prediction of secondary structure was performed using the PSIREP program and the 3D modelling was carried out using the PYRE2 program.

3. Results

3.1. Evaluation of physicochemical parameters in sampling stations

The atmospheric and surface water temperature varied from 29.40 $^{\circ}C$ to 33.10 $^{\circ}C$ and 27.60 $^{\circ}C$ to 31.20 $^{\circ}C$, respectively during low and high tide. The relative humidity of ambient atmosphere reached about 50% to 80%. The annual rainfall during the sampling seasons of this study was about 3815 mm and was prolonged from May to December 2018. Maximum precipitations were recorded during south-west monsoon (May to October) and north-east monsoon (November to December). The salinity and pH were in the range of 31.40 to 33.08 PSU and 8.18 to 8.38 units during low and high tide. Total suspended solids (TSS) concentration widely fluctuated from 9.3 to 48.9 mg/L and the DO range recorded was 6.82 to 7.45 mg/L.

3.2. Population density of *Salmonella*

A total of 10 water and 05 sediment samples were collected along the five selected stations of Port Blair Bays. The graph (Fig. 2) clearly indicates the impact of tidal influences on *Salmonella* population density in the five stations of Port Blair Bay as per sampling period. An increasing trend of *Salmonella* population was observed towards the inner bays with reduction in number towards the outer bays. However, the *Salmonella* population density was substantially increased in PB and JB stations. Throughout the seasonal samplings, the high tide samples scored maximum *Salmonella* count than low tide samples and the faecal indicator bacterial counts highly fluctuated among the sampling sites. The maximum indicator bacterial density was observed in PB station, and the minimum was recorded in MB station.

In FB, MB, JB, PB and AB stations the indicator bacterial counts were recorded 7.80×10^3 , 6.93×10^2 , 3.48×10^5 , 2.35×10^5 and

1.18×10^3 CFU/100 mL during low tide and 2.05×10^3 , 2.36×10^2 , 5.70×10^5 , 5.85×10^5 and 7.51×10^2 CFU/100 mL during high tide, respectively. In sediment samples the higher range of population density (3.80×10^3 , 5.37×10^3 , 4.46×10^5 , 2.83×10^5 and 2.01×10^3 CFU/g) was recorded in all the sampling stations of the Port Blair Bays. The population density of indicator bacteria in low tide was ranged between 6.93×10^2 and 3.48×10^5 CFU/100 mL and in high tide it was ranged between 6.93×10^2 and 5.85×10^5 CFU/100 mL, respectively. The maximum population density of indicator bacteria in water samples was recorded in Phoenix Bay during high tide and minimum was recorded in Minnie Bay water sample during high tide. The *Salmonella* counts in the bottom sediments during low tide ranged between 2.01×10^3 and 4.46×10^5 CFU/g. Maximum counts of *Salmonella* were recorded in Junglighat Bay and minimum count was recorded in Aberdeen Bay sediment sample. The *Salmonella* count was found to be exceeding the USEPA and CPCB limits in all study stations of Port Blair Bays (USEPA: 35 CFU/100 mL for marine water bathing beaches and 61 CFU/100 mL for boat landing centres and harbours) (USEPA, 2003). Faecal coliform standard for bathing, contact water sports and commercial fishing centres (Class SW-II) by Central Pollution Control Board (CPCB), India is 100/100 mL (CPCB, 1995).

3.3. Diversity of *Salmonella* serovars in Port Blair Bays

Based on the characteristic features of *Salmonella*, 1245 presumptive dark purple colour colonies from HiCrome *Salmonella* agar plates were selected for the characterization and species diversity studies from the samples of five stations of (Flat Bay (FB), Minnie Bay (MB), Junglighat Bay (JB) Phoenix Bay and Aberdeen Bay (AB)) Port Blair Bays. Of 1245 *Salmonella* isolates, maximum number was recorded from Junglighat Bay 565 (45.38%) than the other stations (Minnie Bay 60 (04.81%), Flat Bay 117 (09.39%), Phoenix Bay 345 (27.71%), and Aberdeen Bay 158 (12.69%). Based on the morphological, biochemical and phylogenetic analysis; out of 1245 *Salmonella* isolates, 133 isolates were confirmed as the *Salmonella* spp. from the Port Blair Bays (Fig. 3(a)). Of 133 positive isolates, maximum number was recorded from Phoenix Bay 82 (61.65%) than other selected stations (Flat Bay 05 (3.75%) and Junglighat Bay 46 (34.58%). All the presumptive *Salmonella* isolates from the Minnie Bay ($n = 60$) and Aberdeen Bay ($n = 158$) were indicated negative in MacConkey agar and those isolates from the Minnie Bay ($n = 60$) and Aberdeen Bay ($n = 158$) were concluded as false positives and further those stations were not revealed any *Salmonella* contamination in the bay waters. Different ranges of growth were recorded in 133 *Salmonella* isolates, depends on the temperature level. All isolates exhibited good growth in the temperature range of 25 °C–30 °C and 18.04% of the isolates were found grown at 50 °C. Of the total isolates, 0.75% was observed with moderate and fair growth at 60 °C. All isolates revealed negative growth in the temperature range of 70 °C–80 °C (Fig. 3(b)). All isolates revealed acceptable pH tolerance growth at different ranges of pH 7.0 to pH 12.0 (Fig. 3(b)). Among the total isolates, only 15.78% revealed positive growth on all pH ranges. This depicts their tolerance to grow in acidic and alkaliphilic conditions. At pH 4.0, 46.61% isolates revealed negative result and all the isolates were revealed positive growth in the pH range of 6.0–14.0. Isolates from Flat Bay exhibited excellent growth at the pH range of 6.0–12.0, and the isolates from the Junglighat Bay and Phoenix Bay revealed optimum growth at the pH range of 7.0–10.0. All 133 isolates revealed positive results between 0.00 and 3.0% NaCl concentration. It was also observed that, 43 (32.33%) isolates revealed good growth at 6.0% sodium chloride, whereas 13 (9.77%) showed moderate growth and 77 (57.89%) disclosed negative result in the medium. In 10.0% NaCl concentration, 05 (3.76%) showed positive and 128 (96.24%) presented negative result. All these 5 high salt tolerant *Salmonella* isolates were isolated from the Phoenix Bay high tide ($n = 03$), low tide samples respectively. In 15% and 20% NaCl concentration, *Salmonella* isolates were not able to tolerate and no growth were detected in the higher salt concentrations (Fig. 3(b)).

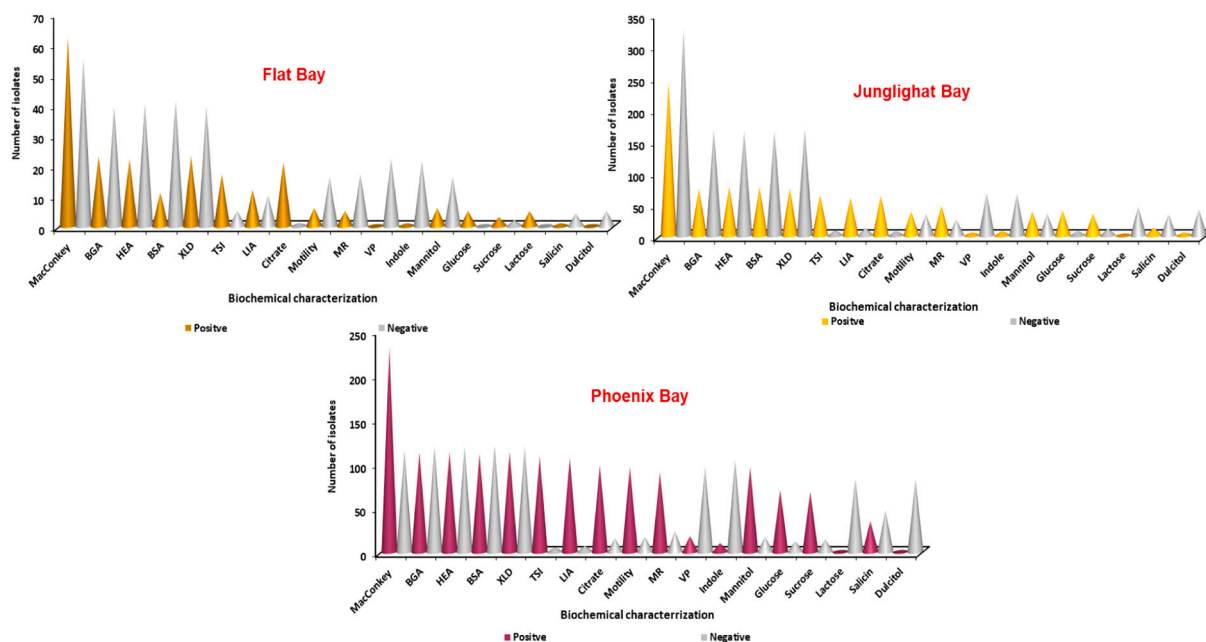
Genus and serovar level confirmation was carried out based on morphological, biochemical and phylogenetic analysis of 16S rDNA sequences (Fig. 4(a–c)) and the isolates were categorized under 21 serovars i.e. *Salmonella enterica* Concord, *Salmonella enterica* Newlands, *Salmonella enterica* Agona, *Salmonella enterica* Heidelberg, *Salmonella enterica* Montevideo, *Salmonella enterica* Typhimurium, *Salmonella enterica* Meleagridis, *Salmonella enterica* Enteritidis, *Salmonella enterica* Shubra, *Salmonella enterica* Indiana, *Salmonella enterica* Paratyphi B, *Salmonella enterica* Paratyphi C, *Salmonella enterica* Paratyphi, Typhi, *Salmonella enterica* Litchfield, *Salmonella enterica* Altendorf, *Salmonella enterica* Rissen, *Salmonella enterica* Stanley, *Salmonella enterica* Infantis, *Salmonella enterica* Senftenberg, *Salmonella enterica* Anatum, *Salmonella enterica* Colombia and unknown serovars. 16S rDNA sequences generated in this study were deposited in GenBank. *Salmonella enterica* Typhimurium and *Salmonella enterica* Enteritidis are predominant throughout the sampling stations of the Port Blair bays. *Salmonella enterica* Newlands also frequently found all the stations, nonetheless in Phoenix Bay station the species diversity was higher than the other stations. *Salmonella enterica* Heidelberg, *Salmonella enterica* Paratyphi B, and *Salmonella enterica* Paratyphi C were also frequently recorded at the different stations. Among the 46 *Salmonella* isolates from the JB station Enteritidis was the dominant serovar, out of 82 *Salmonella* isolates from the PB station Typhimurium was the dominant serovar and in FB station, Newlands serovar was dominating in the diversity (Table 2).

3.4. Antimicrobial resistance profile

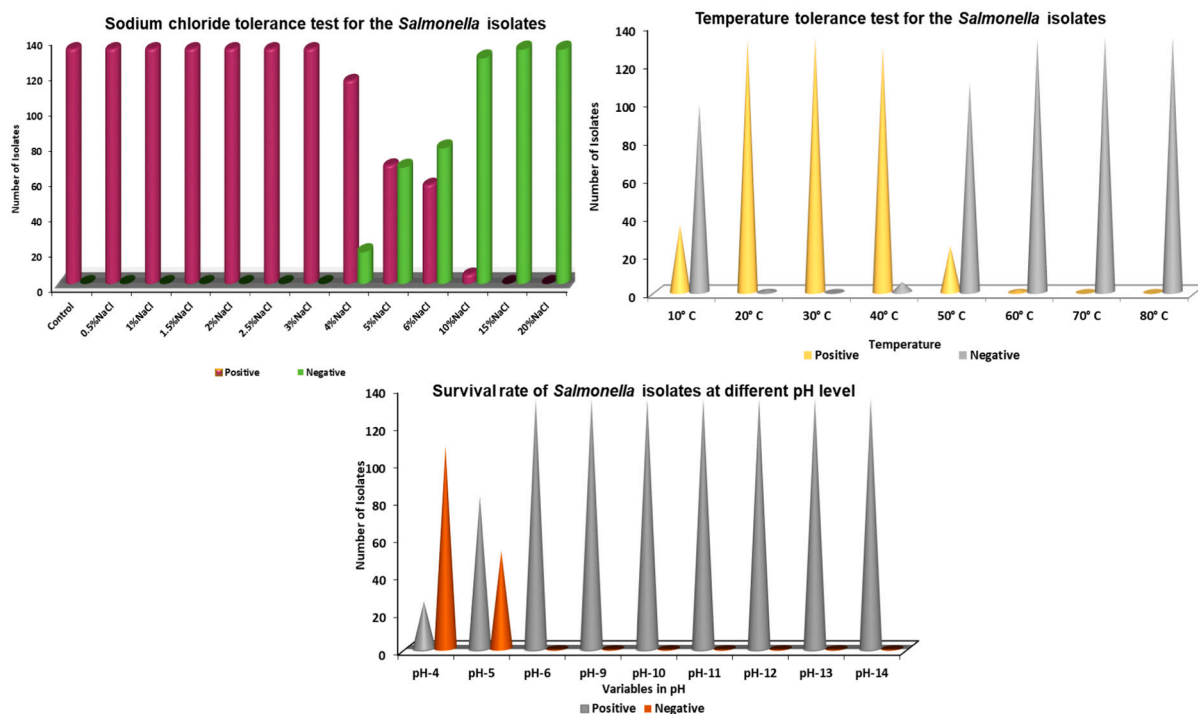
The results of antibiotic susceptibility assay with 133 *Salmonella* isolates from Port Blair Bays against 10 antimicrobial agents are represented in Fig. 5(a). Resistance was observed for all antibiotics tested in this study, and the highest prevalence recorded was against AM (99.24%). *Salmonella* isolates were highly resistant to penicillin-G (81.20%), and Clindamycin (78.19%). Importantly, resistance to the erythromycin and rifampicin was up to 54.88% and 54.13%, respectively, while resistance rates to fluorquinolone like streptomycin and chloramphenicol was 8.27%. However, 10.68% of all isolates ($n = 133$), showed a reduced susceptibility to kanamycin and gentamycin ($n = 2$ and $n = 1$). The isolates showed 37 different antimicrobial resistance patterns and the most prevalent resistance type was antibiotic type 13 (resistance to streptomycin, tetracycline, and amoxicillin), accounting for 43.3% of the isolates. Approximately 77.44% of the strains exhibited multidrug resistance phenotypes by demonstrating resistance to antibiotics from at least three different classes. Among the three stations (Flat Bay, Junglighat Bay and Phoenix Bay) the maximum antibiotic resistant strains were isolated from the Phoenix Bay ($n = 82$). Among the 133 isolates from the Port Blair Bays, maximum multidrug resistant *Salmonella* isolates were detected in the Phoenix Bay high tide water samples ($n = 51$) (Fig. 5(b–d)). Thirty-four percent of *Salmonella* isolates demonstrated multiple antimicrobial resistances to at least two antibiotics. The MAR indices were between 0.11 and 0.22. The maximum MAR index attributed from isolates which exhibited the isolates from the Phoenix Bay.

3.5. Detection of biomarker genes

Out of 133 isolates, 84.14% were found to have the flagellin gene (*flhC*), 73.42% isolates possessed *invA* and *invF* gene, that code for repeat in toxin. Further 68.99% of the isolates were positive for *sitC* gene, which codes the accessory colonization factor. Also 46.84% isolates contain *hlyA* gene, which codes for hemagglutinin protease, whereas 32.28% had *sipC* and *sipF* gene, which codes for toxin coregulated pilus. Heat stable enterotoxin gene, *parE* was found in 31.01% isolates. Among all the sampling stations, maximum number of virulence determinants was found in the *Salmonella* isolates from Phoenix Bay and Junglighat Bay stations. The major virulent genes *gyrA*, *spaP* and *parC* were not detected in any one of the isolates throughout the sampling seasons and stations. The *flhC* and *aadA* gene was recorded as the dominant biomarker gene among the isolates.



(a)



(b)

Fig. 3. Physiological and biochemical characterization of salmonella isolates from Port Blair Bays. (a). Biochemical characterization of *Salmonella* Isolates (b). Physiological characterization of *Salmonella* Isolates.

3.6. Phylogenetic analysis of 16S rDNA sequences

Phylogenetic relationships of the *Salmonella* isolates were ascertained based on the 16S rDNA sequence similarity with reported strains using BLAST sequence similarity search. Upon analysis, it was observed that the deduced 16S rDNA sequences of NIOT_SA1 was highly

homologous (96.34%) with reported sequences of *Salmonella enterica* subsp. *enterica* (GenBank no. MK780060.1). Sequence analysis also specified that 16S rDNA sequences of NIOT_SA1 was closely related to the phylogenetic neighbors; Bacterium strain, MK823711.1 and *Enterobacter cloacae* KC990813.1 with sequence similarity of 91% and 88%, respectively. Phylogenetic analysis based on neighbor-joining tree

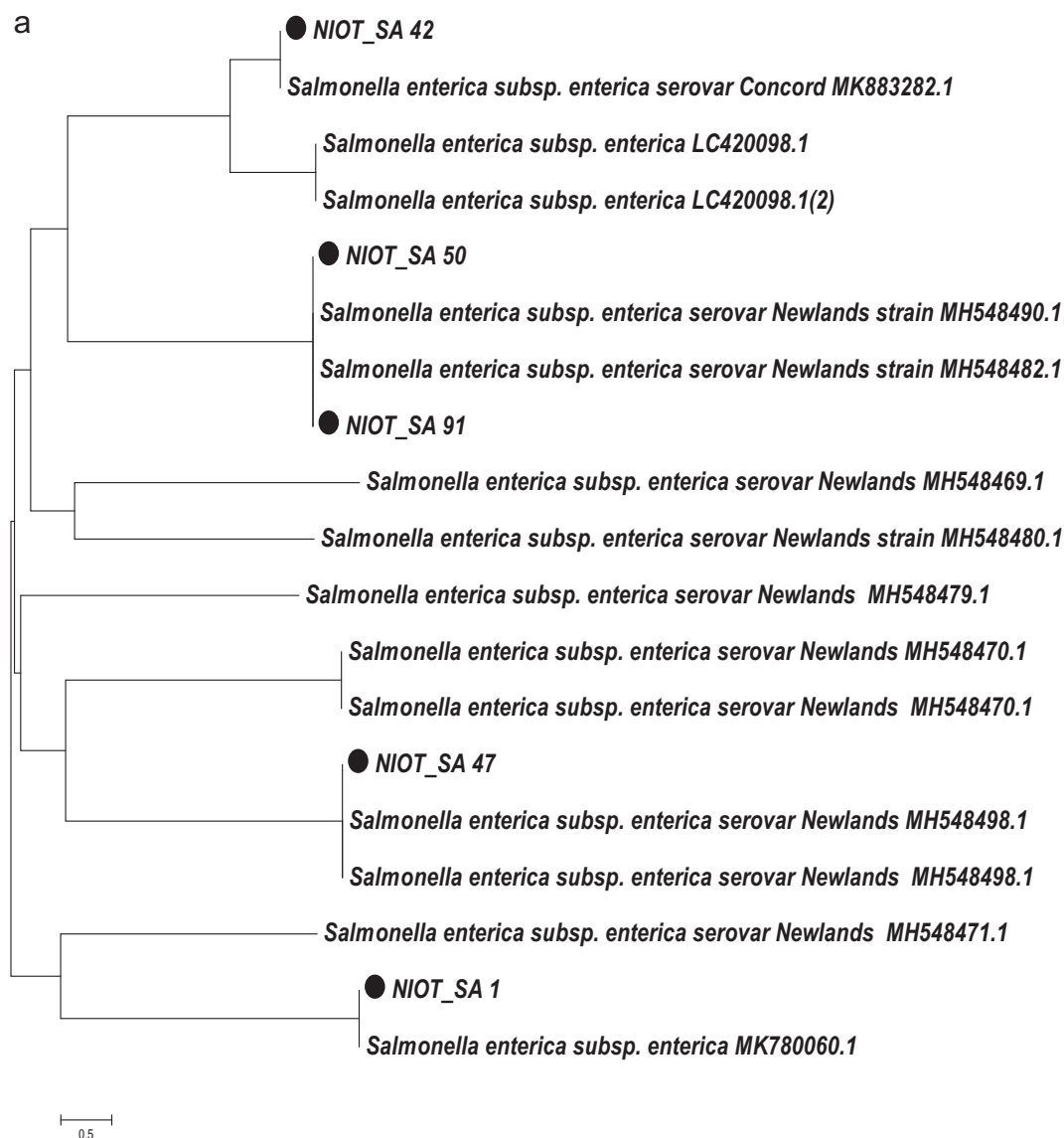


Fig. 4. (a) Neighbor-joining tree based on 16S rDNA sequence data from Flat Bay isolates with sequences available in the GenBank database. Numerical values indicate bootstrap percentage from 1000 replicates. (b) Neighbor-joining tree based on 16S rDNA sequence data from Junghlighat Bay isolates with sequences available in the GenBank database. Numerical values indicate bootstrap percentage from 1000 replicates. (c) Neighbor-joining tree based on 16S rDNA sequence data from Flat Bay isolates with sequences available in the GenBank database. Numerical values indicate bootstrap percentage from 1000 replicates.

(Fig. 4a–c) further revealed that strain NIOT_SA1 formed a distinct branch with *Salmonella enterica subsp. enterica*. The 16S rDNA sequences of NIOT_SA42 were highly homologous (98%) with reported sequences of *Salmonella enterica subsp. enterica* serovar Concord (MK883282.1). Sequence analysis also indicated that 16S rDNA sequence of NIOT_SA42 was highly homologous to the phylogenetic neighbors; *Salmonella enterica subsp. enterica* serovar Paratyphi B (EU118086.1) *Salmonella enterica subsp. enterica* serovar Paratyphi B (EU118082.1) and uncultured bacterium clone (EF604367.1) with sequence similarity of 90% and 89% respectively. Neighbor-joining tree also disclosed that strain NIOT_SA42 forms a single cluster with *Salmonella enterica subsp. enterica* serovar Concord.

The sequences of NIOT_SA47 also established 93% homology with the previous report of *Salmonella enterica subsp. enterica* serovar Newlands (MH548498.1). BLAST analysis also indicated that 16S rDNA sequences of NIOT_SA47 was found extremely related to the phylogenetic neighbors; *Salmonella enterica subsp. enterica* serovar Newlands (MH548469.1) and *Salmonella enterica subsp. enterica* serovar Typhimurium (MH548461.1) with the similarity between 91% and

86% respectively. Neighbor-joining tree (Fig. 4a–c) also disclosed a distinct cluster between NIOT_SA47 and *Salmonella enterica subsp. enterica* serovar Newlands. Upon analysis, it was established that the deduced 16S rDNA sequences of NIOT_SA50 was highly homologous (91.24%) with reported sequences of *Salmonella enterica subsp. enterica* serovar Newlands (MH548490.1). Sequence analysis also specified that 16S rDNA sequences of NIOT_SA50 were closely related to the phylogenetic neighbors; *Salmonella enterica subsp. enterica* serovar Anatum (DQ344534.1), and *Salmonella enterica subsp. enterica* serovar Newlands (MH548469.1) with sequence similarity of 85% and 81%, respectively. Phylogenetic analysis based on neighbor-joining tree (Fig. 4a–c) further revealed that strain NIOT_SA50 formed a distinct branch with *Salmonella enterica subsp. enterica* serovar Newlands.

The sequences of NIOT_SA91 also established 98% homology with the previous report of *Salmonella enterica subsp. enterica* (LC420098.1). BLAST analysis also indicated that 16S rDNA sequences of NIOT_SA91 was found extremely related to the phylogenetic neighbors; *Salmonella enterica subsp. enterica* serovar Typhimurium (DQ344533.1) and *Salmonella enterica subsp. enterica* serovar Anatum (DQ344534.1) with

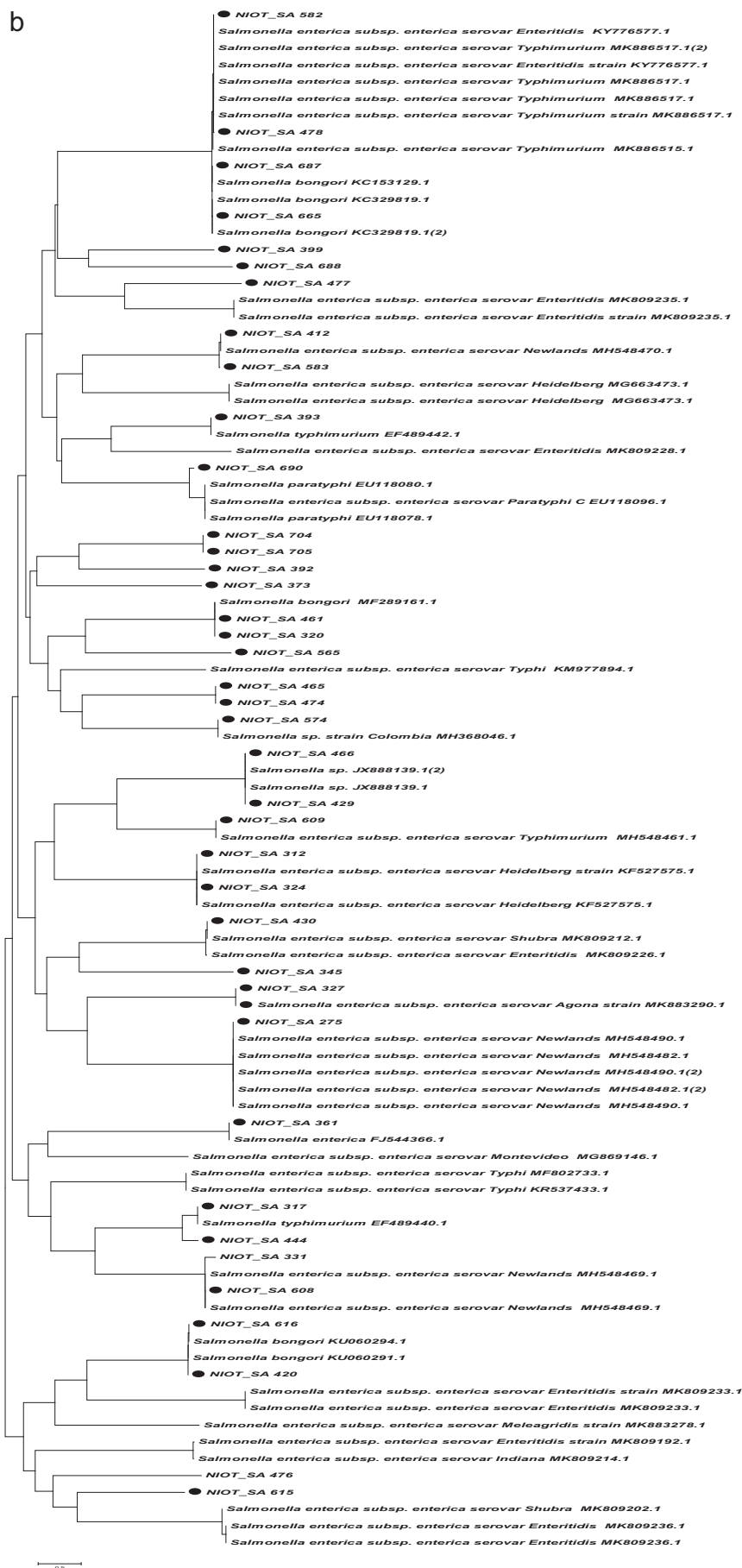


Fig. 4. (continued)

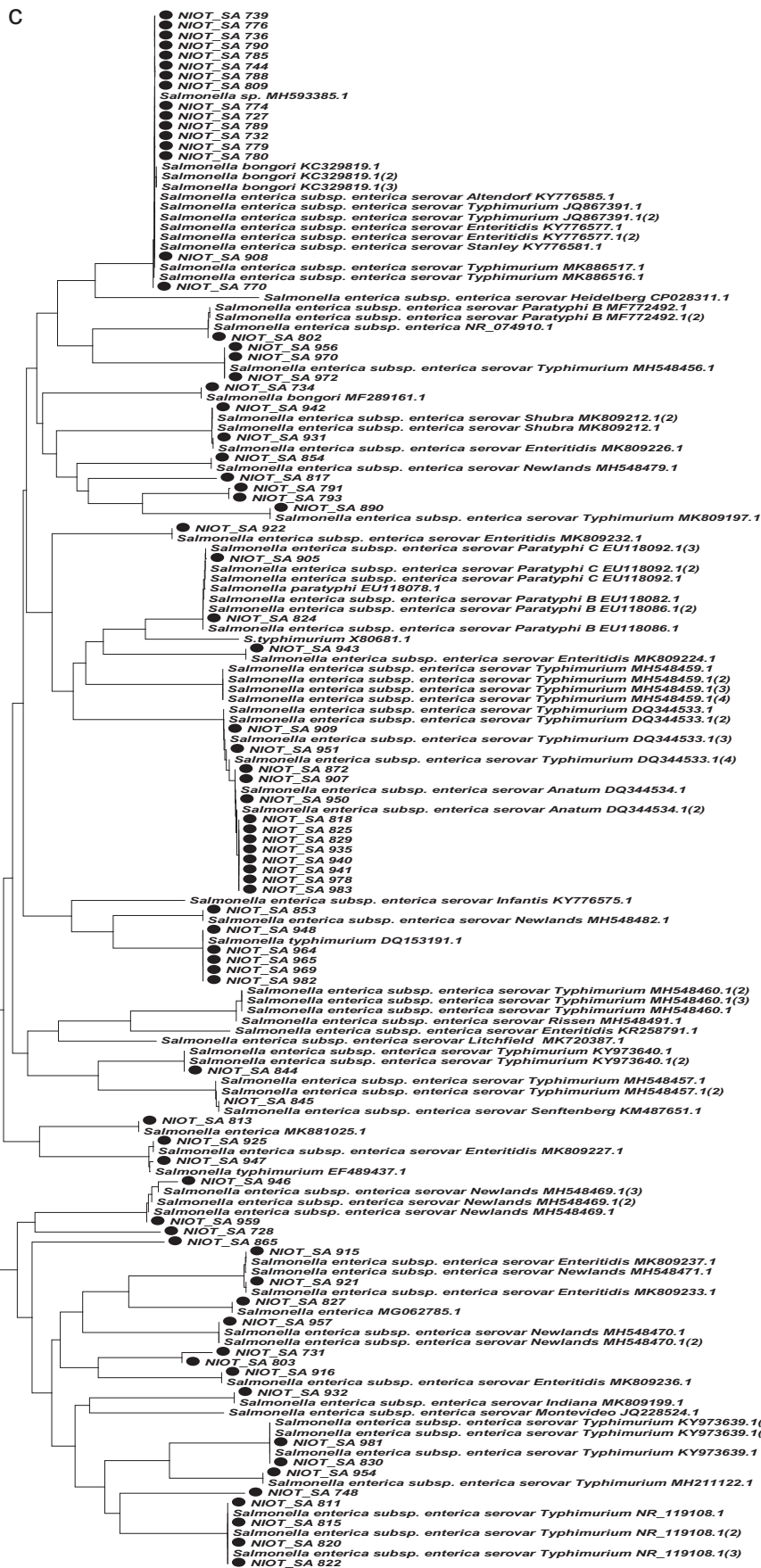


Fig. 4. (continued)

Table 2
Diversity of *Salmonella* serovars in different stations of Port Blair Bays.

Station code	No. of isolates	Salmonella serovar distribution										
		Positive isolates		Concord	Newlands	Agona	Heidelberg	Montevideo	Typhimurium	Meleagridis	Enteritidis	Shubra
FB	117	5	1	3	0	0	0	0	0	0	0	0
MB	60	0	-	-	-	-	-	-	-	-	-	-
JB	565	46	0	5	1	4	1	1	6	1	7	2
PB	345	82	0	8	1	1	1	1	30	0	10	1
AB	158	0	-	-	-	-	-	-	-	-	-	-
Total	1245	133	1	16	2	5	2	2	36	1	17	3

Station code	Salmonella serovar distribution										
	Paratyphi B	Paratyphi C	Paratyphi	Typhi	Litchfield	Altendorf	Rissen	Stanley	Infantis	Senftenberg	Anatum
FB	0	0	0	0	0	0	0	0	0	0	0
MB	-	-	-	-	-	-	-	-	-	-	-
JB	0	1	2	0	0	0	0	0	0	0	0
PB	5	3	0	0	1	1	1	1	1	1	2
AB	-	-	-	-	-	-	-	-	-	-	-
Total	5	4	2	2	1	1	1	1	1	1	2

Station code	Salmonella serovar distribution										
	Paratyphi B	Paratyphi C	Paratyphi	Typhi	Litchfield	Altendorf	Rissen	Stanley	Infantis	Senftenberg	Anatum
FB	0	0	0	0	0	0	0	0	0	0	0
MB	-	-	-	-	-	-	-	-	-	-	-
JB	0	1	2	0	0	0	0	0	0	0	0
PB	5	3	0	0	1	1	1	1	1	1	2
AB	-	-	-	-	-	-	-	-	-	-	-
Total	5	4	2	2	1	1	1	1	1	1	2

the similarity between 78% and 62% respectively. Neighbor-joining tree (Fig. 4(a-c)) also disclosed a distinct cluster between NIOT-SA91 and *Salmonella enterica subsp. enterica*.

16S rDNA sequences of NIOT-SA124 were highly homologous (97%) with reported sequences of *Salmonella typhimurium* (EF489442.1). Sequence analysis also indicated that 16S rDNA sequence of NIOT-SA124 was highly homologous to the phylogenetic neighbors; *Salmonella enterica subsp. enterica* serovar Mbandaka (MG869131.1) and *Salmonella enterica subsp. enterica* serovar Agona (MK883292.1) with sequence similarity of 66% and 58%. Neighbor-joining tree also disclosed that strain NIOT-SA124 forms a single cluster with *Salmonella typhimurium* (Fig. 4(a-c)). Of 133 isolates, 128 isolates were confirmed by molecular level identification and based on the phylogenetic analysis the isolates were categorized under the 22 serovars. *Salmonella enterica* Typhimurium, *Salmonella enterica* Enteritidis, *Salmonella enterica* Heidelberg, *Salmonella enterica* Paratyphi B, and *Salmonella enterica* Paratyphi C were the most dominant serovars.

3.7. In silico analysis of biomarker genes in *Salmonella* isolates

In silico sequence analysis of the biomarker gene divulged that, there was a significant level of similarity between the sequences obtained in this study with that of previous reports in the sequence database. The *hilA*, *hilC* and *hilD* genes showed a less significant divergence with the sequences reported from other species. Sequence analysis of *filC* and *invA* genes revealed a significant homology with the previous reports and hence there is an insignificant variation in the phylogenetic tree. In *aadA* gene, it was observed that the *Salmonella* sp. alone formed a separate cluster with that of other sequences. Secondary and 3D structure prediction was carried out for *sitA*, *sitB*, *sitC*, *aadA*, *invA*, *invB* and *filC* genes. Secondary structure of *sitA* was predicted to have maximum of sixteen α -helical structure and nine β -strand with maximum hydrophilic molecules. The *sitB* was predicted with maximum of sixteen α -helical structure and nine β -strand maximum hydrophilic molecules and *sitC* was predicted with maximum of sixteen α -helical structure and nine β -strand minimums of hydrophilic molecules. Secondary structure prediction analysis of *invA* and *invF* also revealed to have twelve α -helical structure and seven β -strand with maximum hydrophilic molecules, and with twelve α -helical structure and seven β -strand with maximum hydrophilic molecules respectively. The 2D structure analysis for the *filC* gene was revealed with maximum of sixteen α -helical structure and nine β -strand minimums of hydrophilic molecules but the *aadA* gene revealed maximum of β -strand with maximum hydrophobic molecules. Large number of regions was also predicted for favourable formations of coils in the deduced amino acid sequences of both strains.

The 3D structural models of *sitA*, *sitB* and *sitC* were compatible with the secondary structure prediction analysis. Three dimensional studies on *sitA* protein of *Salmonella* NIOT-SA983 revealed 47% of α -helix and 16% of β -strand. 35% of α -helix and 26% of β -strand was observed in *sitB* protein and 47% of α -helix and 16% of β -strand for the *sitC* protein. Prediction results displayed that, the alpha helices are mainly present in C-terminal than the middle and N-terminal regions, which appear in many members of transferase family. Homology analysis of marker gene proteins with Protein Data Base (PDB) templates: Maximum of 43% identity and minimum 15% identity was observed in *filC* protein and maximum 43% and minimum 13% identity was observed in *aadA* protein of *Salmonella* NIOT-SA983 (Fig. 6).

4. Discussion

The results of this study indicate is responsible for the tidal variations in the prevalence of *Salmonella* in Port Blair Bays, with an overall incidence of 10.68%. *Salmonella* have been reported to be considerably more prevalent on the coasts of Portugal and the Northeast coast of Spain, where values for the incidence of *Salmonellae* in seawater was

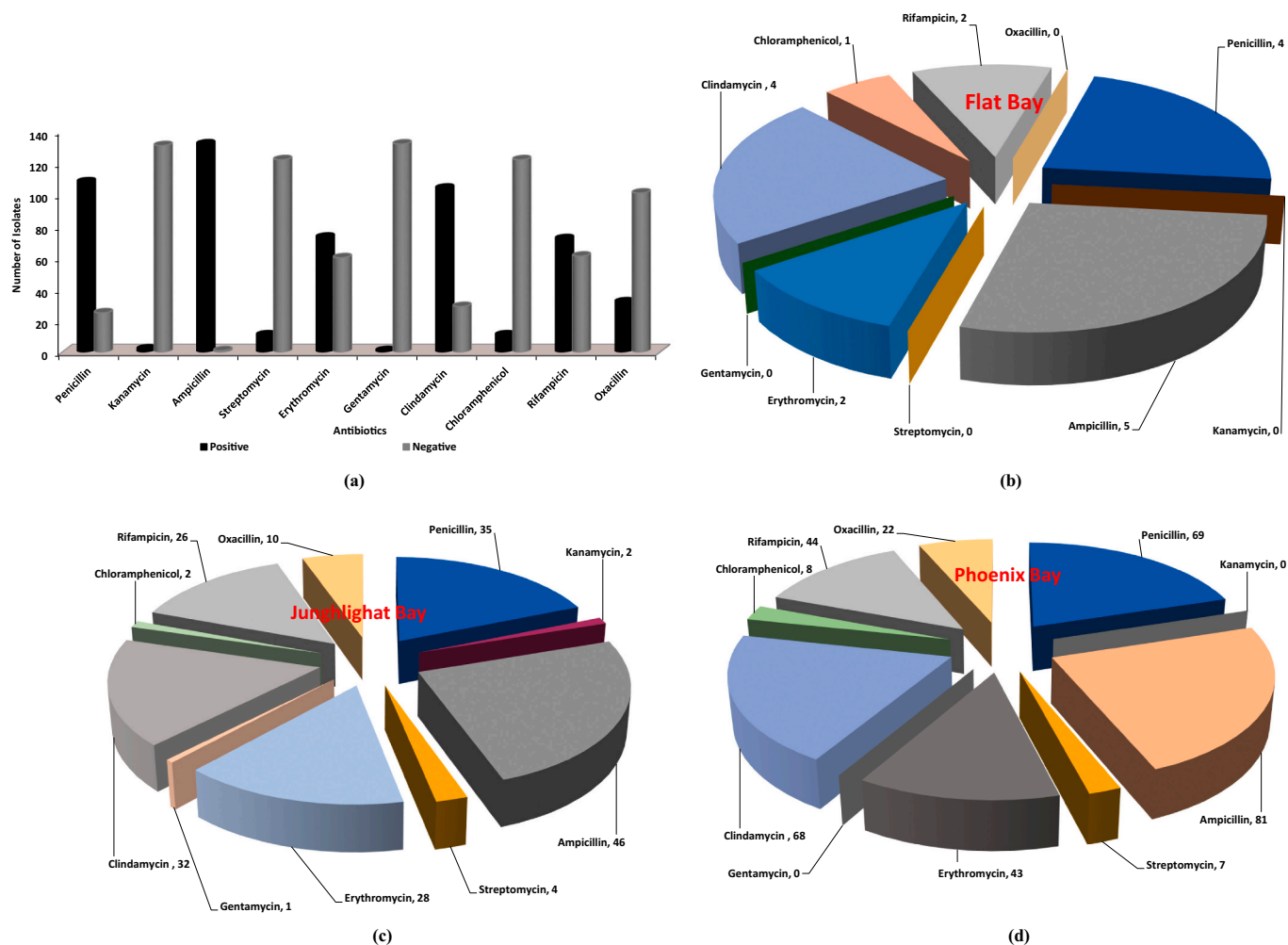


Fig. 5. (a) Antibiotic resistant and sensitive pattern of *Salmonella* isolates (b) Antibiotic resistant pattern of *Salmonella* isolates from Flat Bay (c) Antibiotic resistant pattern of *Salmonella* isolates from Junghlight Bay (d) Antibiotic resistant pattern of *Salmonella* isolates from Phoenix Bay.

41%, 72% (Dionisio et al., 2000), and 26% in Spain (Polo et al., 1999) respectively were reported. The *Salmonella* diversity in present study as reflected in the identification of 22 serovars among 133 strains of *Salmonella* is typical of what was found during previous studies. In a study by Willson and Moore (1996), 17 different serovars were identified among the 37 strains isolated, while a study in Portugal found 17 different serovars among 45 *Salmonella* isolates (Dionisio et al., 2000). In a study of coastal waters in Japan, 18 different serovars were identified among 251 strains (Venkateswaran et al., 1989) and a total of 197 isolates representing 13 different serotypes, were recovered from Suwannee River Basin of the coastal plain of South Georgia (Bradd et al., 2009). The serovars in present study were notably different than those of most frequently identified in clinical, animal, and food samples in India. According to the data from the Spanish Weekly Epidemiological Bulletin, between 1996 and 1999 the five *Salmonella* serovars most frequently isolated from clinical samples were Enteritidis, Typhimurium, Hadar, Virchow, and Brandenburg (Dionisio et al., 2000). Isolates from animal samples were more varied, and the predominant serovars were Typhimurium, Enteritidis, and Abortusovis. In environmental samples, serovars Enteritidis, Typhimurium, and Hadar were identified most often (Polo et al., 1999). These frequency patterns of *Salmonella* serovar isolates differ notably from the data obtained in present study, where serovars Typhimurium, Newlands and Enteritidis made up 51.88% of the isolates. The discrepancy between serovars originating from clinical or animal sources and those from the marine environment has been observed in previous studies of the presence of

Salmonellae in coastal water (Dionisio et al., 2000) and seafood products (Heinitz et al., 2000). Assuming that the main source of *Salmonella* contamination in the marine environment is of anthropogenic or animal origin, the different population structures may be attributed to the different rates of growth and survival of these serovars in the marine environment. *Salmonella* serovar Typhimurium is the clinically important serovar found most often in this study and in most of the previous studies of *Salmonella* contamination in marine environments (Dionisio et al., 2000; Heinitz et al., 2000), which attests to its capacity of adaptation and survival in this environment, as reported previously (Baudart et al., 2000a,b). This resistance of *Salmonella* serovar Typhimurium has also been demonstrated in experimental studies (Galdiero et al., 1994). Most *Salmonella* survival studies have been conducted under controlled and stable conditions, and little is known about the survival of most *Salmonella* serovars in constantly changing and unpredictable marine ecosystems. *Salmonella* serovar Typhimurium was detected most frequently in this study (27.06% of the isolates); followed by serovar Enteritidis (12.78% of the isolates). Previous research studies indicate that the presence of serovar Typhimurium and Enteritidis is increasing in coastal marine environments. It was one of the predominant serovars in the coastal waters of Portugal (Dionisio et al., 2000), in crustaceans from India (Hatha and Lakshmanaperumalsamy, 1997), in seafood imported to the United States especially from tropical countries (Heinitz et al., 2000), and in environmental samples in France and Brazil (Baudart et al., 2000a,b). The Spanish Weekly Epidemiological Bulletin (Usera et al., 1998, 1999) reports that the number of

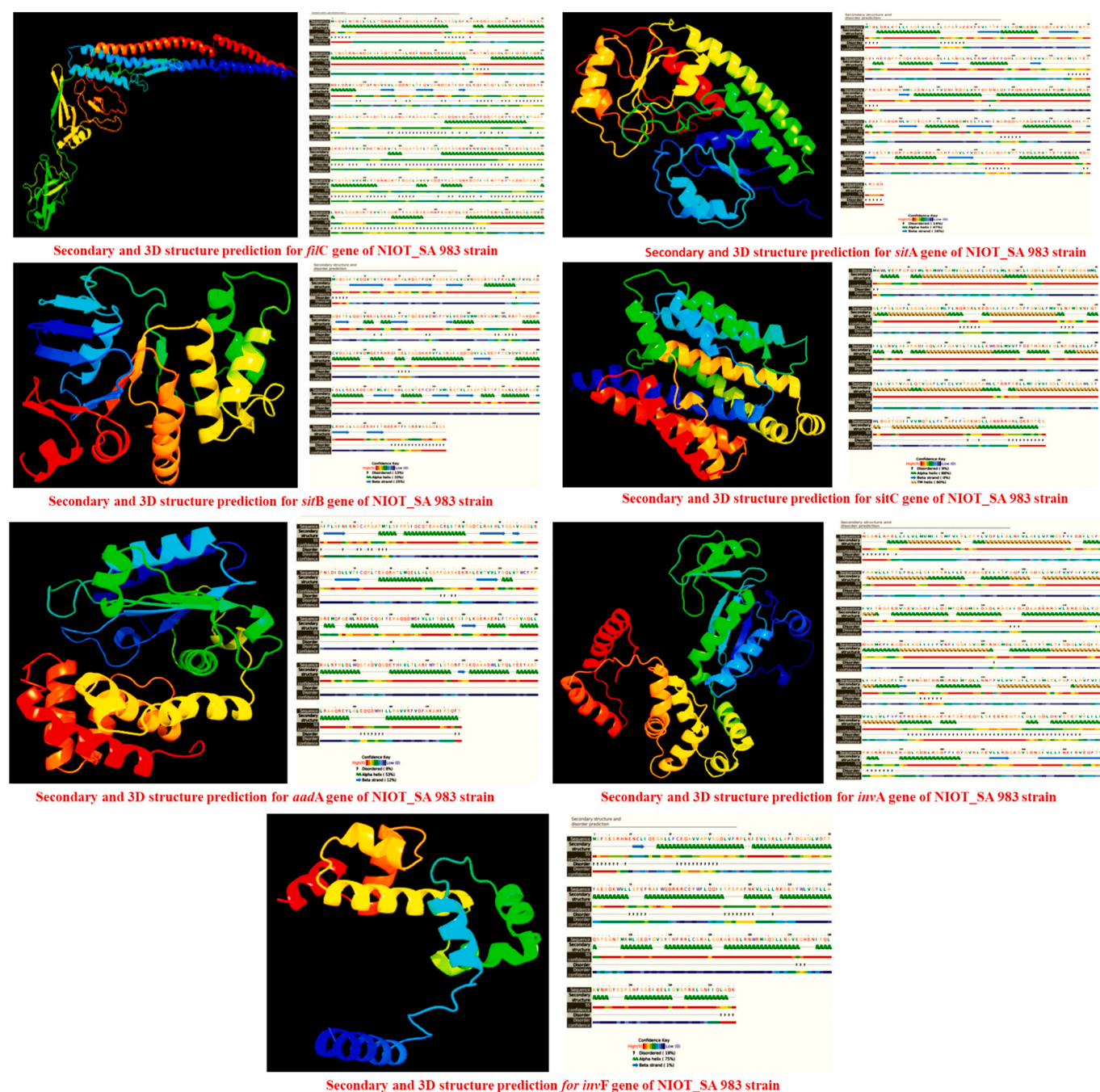


Fig. 6. Structure prediction of biomarker genes in the *Salmonella* isolates.

Salmonella serovar Typhimurium isolates from food and animal feed increased from 23% in 1997 to 89% in 1998, and during that period, serovar Enteritidis was the third most frequently isolated serovar.

Worldwide, *Salmonella* is recognized as a major gastrointestinal pathogen for humans and animals, whereas, *Salmonella typhimurium* and *Salmonella enteritidis* were considered as the major serovars that can be infecting human after eating food of animal origin, but this data may be different over the years and countries. In this study, the antimicrobial resistance analysis revealed the presence of *Salmonella* Heidelberg resistant to quinolones and *Salmonella* Typhimurium resistant to multiple antibiotics including third generation cephalosporins. Other studies showed that the resistance to quinolones, fluoroquinolones, and third generation cephalosporins in *Salmonella* has recently increased (Karraouan et al., 2017). This resistance may be due to the intensive use of antibiotics in the veterinary and human health

care by contributing to the acquisition of resistance genes and exerting a selective pressure for the development of resistant bacteria (Foley and Lynne, 2008) and may change depending on serotype, time, source, and geographic region of the isolate (Acar et al., 2017). The pathogenicity of *Salmonella* is influenced by various factors encoded by different virulence genes that play an important role in different steps of pathogenicity, such as adhesion, invasion, intracellular survival, systemic infection, toxin production, and iron acquisition (Skyberg et al., 2006). In this study, all isolates contain the six major virulent genes (*sitA*, *sitC*, *sitB*, *filC*, *aadA*, and *invF*) out of sixteen tested virulent genes. The host cell invasion and intracellular survival of *Salmonella* depends on several genes that are clustered on pathogenicity islands (PIs), and these effector proteins are transferred by two type III secretion systems, namely, T3SS1 and T3SS2. Moreover, *Salmonella* strains expresses several invasion genes located on the *Salmonella* pathogenicity islands 1

and 2 (SPI-1 and SPI-2), which promote bacterial uptake and invasion of non-phagocytic cells to cross the epithelial barrier (Bhunia, 2018). Among these genes, *filC* is required for invasion and secretion system (Klein et al., 2000). Thus, the *sitB* gene is responsible for the entry into non-phagocytic cells and killing the macrophages. It has been shown that the targeting of *sitC* by the Inv-Spa type III secretion apparatus is necessary and sufficient for the induction of macrophage apoptosis (Hersh et al., 1999). The *invF* gene is implicated in iron acquisition (Baumler et al., 1998). However, the *sitC* gene encodes an important transporter of iron. Also, a study carried out by Boyer et al. (2002) on *Salmonella typhimurium* revealed that *sitABCD* is required for the replication of *Salmonella* inside macrophages and for the creation of virulence in susceptible animals (Boyer et al., 2002). On the other hand, the *sifA* gene is required for the formation of filamentous structures in the lysosome vacuoles of infected epithelial cells (Stein et al., 1996). A previous study conducted on *Salmonella typhimurium* showed that *spiA* gene codes for an outer membrane component that is essential for virulence in host cells (Ochman et al., 1996). Another study carried out by Dong et al. (2001) revealed that *spiA* gene is associated with biofilm formation mechanism. It has been demonstrated that biofilm formation improves the ability of microorganisms to withstand stresses, such as desiccation, temperature extremes, antibiotics, and antiseptics (Marin et al., 2009), which allowed these bacteria to survive longer in animal farms and to contaminate meats and eggs, which remain the main vehicles of *Salmonella* transmission to humans. Presence of the virulence factors in *Salmonella* isolates from this study indicates the public health risk of coastal waters in Port Blair Bays. Most of the countries it is a challenge for laboratories to systematically identify *Salmonella* isolates on the basis of biochemical and serological identification using Kauffmann-White scheme because of the cost of reagents and the time consumption for the results. In this study, we conducted a rapid and cost-effective method to identify the isolated *Salmonella* strains using DNA sequencing tools. Furthermore, the sequence analysis of the 16S rDNA is being regularly used to accurately identify bacterial species and to perform the taxonomic studies in clinical and scientific investigations (Petti et al., 2005). However, the phylogenetic analyses can be used to understand the emergence of pathogenic bacteria and to study the relationship between isolated strains and other reference strains, implicated in animal or human diseases (Ed-Dra et al., 2018). In this study, the analysis of phylogenetic tree disclosed the presence of different 16S rDNA profiles among the isolated *Salmonella* serovars, suggesting the presence of different contamination sources due to the diversity of pollutants and the tidal influences in Port Blair Bays. The results obtained in this study showed that the sequencing of 16S rDNA amplicons was a suitable tool for identification of *Salmonella* strains at the genus and species level. In addition, the phylogenetic analysis revealed a high diversity in all clusters of this bacterium. Furthermore, the antimicrobial resistance and the virulent gene analysis suggest that coastal waters of Port Blair Bays are contaminated with *Salmonella* strains with a major risk to public health.

This study constitutes the first report on the occurrence of multidrug resistant *Salmonella* serovars in the waters of the Port Blair Bays, providing insights into the prevalence, bacterial species affiliation and molecular epidemiology of multidrug resistant strains (MDRs). Herein, we identified *Salmonella* species carrying clinically important virulence factors determinants that have not been previously isolated from natural environments. Our data highlight the public health problem arising from the presence of multidrug resistant *Salmonella* species in polluted bays and also beach waters designated for recreational uses, which have the potential to ultimately act as mediators in the gene flow between marine coastal areas and clinical settings. This study also reveals the importance of regular monitoring of coastal water quality, need for proper regulation of municipal waste water discharge in the coastal areas, continuous monitoring for the presence of pathogens in water, sediment, fin and shell fishes. This work also demonstrates the importance of environmental studies seasonably for further understanding

the evolution and epidemiology of antibiotic resistance in order to reduce or prevent its public health and ecological consequences.

Authors' contribution

The research concept and the experiments were executed by BM, LA and KS. NVV and GD analysed the data and reviewed the manuscript. All of the authors approved the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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